



Vishal S R 250701844-2025-CSE ▾

V2**Started on** Sunday, 30 August 2026, 12:42 AM**State** Finished**Completed on** Sunday, 30 August 2026, 12:43 AM**Time taken** 1 min 18 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of $\text{arr}[i] * i$, where i is the index of the element ($i = 0, 1, 2, \dots, N$). Write an algorithm based on Greedy technique with a Complexity $O(n \log n)$.

Input Format:

First line specifies the number of elements- n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int compare(const void *a, const void *b)
5  {
6      return (*(int *)a - *(int *)b);
7  }
8
9  int main()
10 {
11     int n;
12     scanf("%d", &n);
13
14     int arr[n];
15
16     for (int i = 0; i < n; i++)
17     {
18         scanf("%d", &arr[i]);
19     }
20
21
22     qsort(arr, n, sizeof(int), compare);
23
24     long long sum = 0;
25
26
27     for (int i = 0; i < n; i++)
28     {
29         sum += (long long)arr[i] * i;
30     }
31
32     printf("%lld", sum);
33
34     return 0;
35 }
```

	Input	Expected	Got	
✓	5	40	40	✓
	2			
	5			
	3			
	4			
	0			

	Input	Expected	Got	
✓	10 2 2 2 4 4 3 3 5 5 5	191	191	✓
✓	2 45 3	45	45	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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